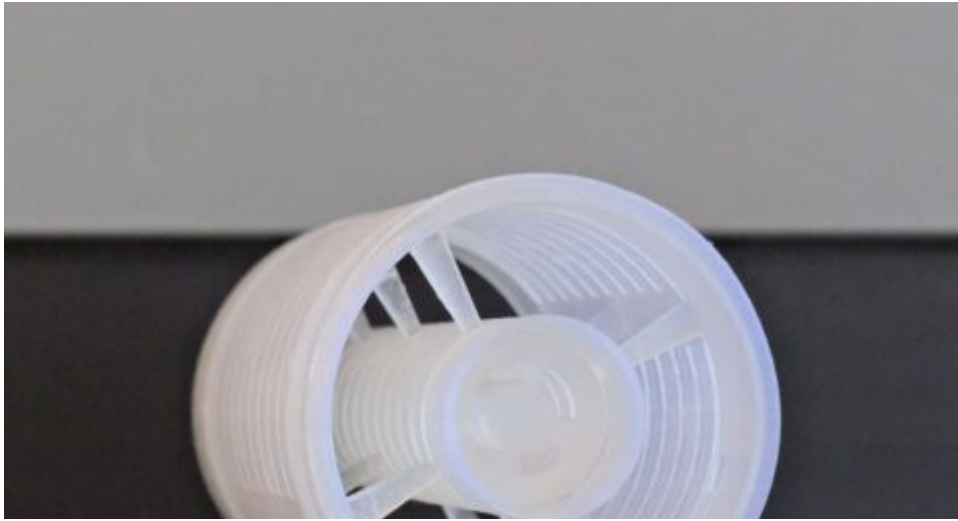


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ACOUSTIC

CONCEPT

SMARTER SOUND SHIELD BLOCKS NOISE WITHOUT BLOCKING AIR

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15 August 2025

A new development in sound-silencing materials from [Boston University](#) could offer promising solutions for designers seeking quieter, healthier environments—without compromising on ventilation. Researchers from the Zhang Lab have unveiled a new generation of ultra-open acoustic metamaterials that significantly reduce a wide range of noises while allowing air to pass freely.

**ACOUSTIC
PERFORMANCE MEETS
AIRFLOW**

Originally introduced in 2019, the lab's Acoustic Metamaterial Silencer aimed to block specific sound frequencies in applications like fans, HVAC systems, and mechanical equipment. This early version leveraged the physical phenomenon of Fano resonance to suppress narrowband noise while maintaining open air channels—crucial for systems that require ventilation.

Now, Professor Xin Zhang and her team have expanded this concept to address broader and more

Reports, presents the Phase Gradient Ultra-Open Metamaterial (PGUOM)—a novel structure that offers broadband acoustic silencing, capable of muffling a wide spectrum of sound frequencies in real time, even as they change in pitch and volume.

DESIGN IMPLICATIONS FOR BUILT ENVIRONMENTS

This advancement holds major implications for architectural and interior applications, especially in environments where both noise control and airflow are essential, such as offices, factories, public transit hubs, and open-plan buildings. Unlike traditional sound barriers, PGUOM functions more like noise-canceling headphones, intelligently managing acoustic energy without sealing off airflow. This makes it particularly relevant for architects and engineers designing ventilated façades, acoustic ceilings, or mechanical systems where maintaining fresh air and sound comfort are both priorities.

Beyond performance, the concept of a highly open, passive acoustic solution aligns well with broader goals of sustainability and energy efficiency. By reducing the need for powered acoustic systems or energy-intensive ventilation alternatives, PGUOM offers a low-maintenance, material-efficient option that supports both occupant comfort and environmental performance. While the current research does not explicitly focus on biobased or recycled materials, the form and function of the metamaterial open new design possibilities for eco-conscious product development in acoustic materials.

This technology marks a step forward in material innovation for the built environment, where multifunctional performance—silence, breathability, and efficiency—is becoming increasingly essential.

Source & photo: Boston University College of Engineering

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